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Electrical track assembly

Abstract

A support member for an electrical track assembly comprises a first tracking including a first conductor and a second tracking including a second conductor, the support member comprising: a first support member contact; a second support member contact; an electrical load including a first load contact and a second load contact; and a switch assembly configured for selectively electrically connecting said first conductor with the first load contact, selectively electrically connecting said second conductor with the second load contact, and preventing current flow (i) between said first conductor and the second load contact and (ii) between said second conductor and the first load contact.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. patent application Ser. No. 17/116,959, filed on Dec. 9, 2020, now U.S. Pat. No. 11,807,142, issued Nov. 7, 2023, which is a continuation-in-part application of U.S. patent application Ser. No. 16/294,289, filed on Mar. 6, 2019, now U.S. Pat. No. 11,299,075, issued Apr. 12, 2022, the disclosures of which are hereby incorporated by reference in their entireties as though fully set forth herein.

TECHNICAL FIELD

(1) The present disclosure generally relates to track assemblies, including track assemblies with electrical components and track assemblies that provide and/or facilitate electrical connections.

BACKGROUND

(2) This background description is set forth below for the purpose of providing context only. Therefore, any aspect of this background description, to the extent that it does not otherwise qualify as prior art, is neither expressly nor impliedly admitted as prior art against the instant disclosure.

(3) Some track designs may not be configured for connections in multiple directions/configurations, and/or may require expensive/complex electrical systems.

(4) There is a desire for solutions/options that minimize or eliminate one or more challenges or shortcomings of track assemblies. The foregoing discussion is intended only to illustrate examples of the present field and is not a disavowal of scope.

SUMMARY

(5) In embodiments, an electrical track assembly may include a pair of tracks, including a first track including a first conductor, a second track include a second conductor; and a support member configured for connection with, removal from, and movement along the pair of tracks in a first configuration and a second configuration. The support member may include a first support member contact, a second support member contact, an electrical load including a first load contact and a second load contact, and a manual switch assembly configured to selectively electrically connect the first conductor with the first load contact, selectively electrically connect the second conductor with the second load contact, and prevent current flow (i) between the first conductor and the second load contact and (ii) between the second conductor and the first load contact, the manual switch assembly including a blocking element.

(6) The foregoing and other potential aspects, features, details, utilities, and/or advantages of

examples/embodiments of the present disclosure will be apparent from reading the following description, and from reviewing the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) While the claims are not limited to a specific illustration, an appreciation of various aspects may be gained through a discussion of various examples. The drawings are not necessarily to scale, and certain features may be exaggerated or hidden to better illustrate and explain an innovative aspect of an example. Further, the exemplary illustrations described herein are not exhaustive or otherwise limiting, and are not restricted to the precise form and configuration shown in the drawings or disclosed in the following detailed description. Exemplary illustrations are described in detail by referring to the drawings as follows:

(2) FIG. 1 is a side view generally illustrating an embodiment of an electrical assembly according to teachings of the present disclosure.

(3) FIG. 2A is a cross-sectional view generally illustrating an embodiment of an electrical assembly with a support assembly connected in a first orientation according to teachings of the present disclosure.

(4) FIG. 2B is a cross-sectional view generally illustrating an embodiment of an electrical assembly with a support assembly connected in a second orientation according to teachings of the present disclosure.

(5) FIGS. 3A, 3B, and 3C are schematic views generally illustrating portions of embodiments of an electrical assembly according to teachings of the present disclosure.

(6) FIGS. 4A and 4B are schematic views generally illustrating portions of embodiments of an electrical assembly according to teachings of the present disclosure.

(7) FIGS. 5A and 5B are schematic views generally illustrating portions of embodiments of an electrical assembly according to teachings of the present disclosure.

(8) FIGS. 6A and 6B are schematic views generally illustrating portions of an embodiment of a control circuit an electrical assembly according to teachings of the present disclosure.

(9) FIGS. 7A and 7B are schematic view generally illustrating portions of an embodiment of a control circuit of an electrical assembly according to teachings of the present disclosure.

(10) FIGS. 7C and 7D are schematic views generally illustrating portions of embodiments of a control circuit of an electrical assembly according to teachings of the present disclosure.

(11) FIG. 8 is a side view generally illustrating an embodiment of an electrical track assembly with a first support member in a first configuration and a second support member in a second configuration according to teachings of the present disclosure.

(12) FIG. 9 is an end view generally illustrating an embodiment of an electrical track assembly with a support member in a first configuration according to teachings of the present disclosure.

(13) FIG. 10 is an end view generally illustrating an embodiment of an electrical track assembly with a support member in a second configuration according to teachings of the present disclosure.

(14) FIG. 11 is an end view generally illustrating an embodiment of an electrical track assembly with a support member in a second configuration according to teachings of the present disclosure.

(15) FIG. 12 is a diagram generally illustrating an embodiment of an electrical track assembly with a switch circuit in a first switch circuit position according to teachings of the present disclosure.

(16) FIG. 13 is a diagram generally illustrating an embodiment of an electrical track assembly with a switch circuit in a second switch circuit position according to teachings of the present disclosure.

(17) FIG. 14 is a diagram generally illustrating an embodiment of an electrical track assembly with a switch circuit in a third switch circuit position according to teachings of the present disclosure.

(18) FIG. 15 is a diagram generally illustrating an embodiment of an electrical track assembly with

two support members having switches in first positions according to teachings of the present disclosure.

(19) FIG. **16** is a diagram generally illustrating an embodiment of an electrical track assembly with two support members having switches in second positions according to teachings of the present disclosure.

(20) FIG. **17** is a diagram generally illustrating an embodiment of an electrical track assembly with two support members having switches in a first position and a second position, respectively, according to teachings of the present disclosure.

(21) FIG. **18** is a diagram generally illustrating an embodiment of an electrical track assembly with two support members having switches in a first position and a second position, respectively, according to teachings of the present disclosure.

(22) FIG. **19** is a diagram generally illustrating an embodiment of an electrical track assembly with two pairs of tracks and five support members according to teachings of the present disclosure.

(23) FIG. **20** is an end view generally illustrating an embodiment of an electrical track assembly with a support member in a first configuration according to teachings of the present disclosure.

(24) FIG. **21** is an end view generally illustrating an embodiment of an electrical track assembly with a support member in a second configuration according to teachings of the present disclosure.

(25) FIG. **22** is an enlarged view an embodiment of a switch and portions of an embodiment of a blocking element according to teachings of the present disclosure.

(26) FIGS. **23A-23C** are end views of embodiments of tracks and power sources of electrical track assemblies according to teachings of the present disclosure.

(27) FIGS. **24-28B** are end views generally illustrating embodiments of electrical track assemblies with support members according to teachings of the present disclosure.

(28) FIG. **29** is an end view generally illustrating an embodiment of an electrical track assembly with a support member according to teachings of the present disclosure.

DETAILED DESCRIPTION

(29) Reference will now be made in detail to embodiments of the present disclosure, examples of which are described herein and illustrated in the accompanying drawings. While the present disclosure will be described in conjunction with embodiments and/or examples, they do not limit the present disclosure to these embodiments and/or examples. On the contrary, the present disclosure covers alternatives, modifications, and equivalents.

(30) In embodiments, such as generally illustrated in FIGS. **1**, **2A**, and **2B**, an electrical assembly **20** may include a support assembly **22**, a support member **30**, a track assembly **38**, and/or a control circuit **52**. The control circuit **52** may include at least one of a switch/relay assembly **60**, a diode assembly **80**, and/or a switch assembly **90**. The support member **30** may support and/or be connected to a seat **30A**. The electrical assembly **20** may be configured to control, at least in part, movement of the support assembly **22**. The support member **30** may be configured for selective connection (e.g., electrical and/or mechanical) with the track assembly **38**. For example and without limitation, the support member **30** may be configured to provide electrical connection between the track assembly **38** and a support assembly **22**, which may include a vehicle seat **30A** and/or other elements that may be connected to the support member **30**. The track assembly **38** may be connected to a mounting surface **24**, such as a vehicle floor.

(31) With embodiments, a support member **30** may be connected to and/or be configured to engage a track assembly **38**. The support member **30** and/or the track assembly **38** may extend substantially longitudinally (e.g., in an X-direction). For example and without limitation, the support member **30** may move (e.g., slide, roll, translate, etc.) in a longitudinal direction along the track assembly **38**. The support member **30** may selectively engage and/or disengage from the track assembly **38**. The support member **30** may be inserted into and/or be removed from the track assembly **38** in a Z-direction (e.g., a vertical direction). The support member **30** may, for example and without limitation, include a cassette configuration.

(32) In embodiments, such as generally illustrated in FIG. 2A, a track assembly 38 may include a first track 40 and/or a second track 40'. The first track 40 and/or the second track 40' may extend substantially in a longitudinal direction (e.g., the X-direction). The first track 40 and the second track 40' may be substantially the same and/or may be disposed in a mirrored configuration. The first track 40 may be offset in the Y-direction from the second track 40'. The first track 40 and/or the second track 40' may include substantially the same length. The support member 30 may include a first portion 32 and/or a second portion 34. The first portion 32 and/or the second portion 34 may selectively engage the first track 40 and/or the second track 40'. For example and without limitation, the first portion 32 of the support member 30 may engage the first track 40 and/or the second portion 34 of the support member 30 may engage the second track 40' (e.g., a forward facing support assembly 22), and/or the first portion 32 of the support member 30 may engage the second track 40' and/or the second portion 34 of the support member 30 may engage the first track 40 (e.g., a rearward facing support assembly 22).

(33) With embodiments, the first track 40 and/or the second track 40' may include an outer track 42, 42' and/or an inner track 44, 44'. The outer tracks 42, 42' may include a first/bottom wall 42.sub.1, 42.sub.1', a second wall 42.sub.2, 42.sub.2', and/or a third wall 42.sub.3, 42.sub.3'. The bottom wall 42.sub.1, 42.sub.1', the second wall 42.sub.2, 42.sub.2', and/or the third wall 42.sub.3, 42.sub.3' may be connected to form a generally U-shaped configuration. The bottom wall 42.sub.1, 42.sub.1' may, for example, be substantially planar. The second wall 42.sub.2, 42.sub.2', and/or the third wall 42.sub.3, 42.sub.3' may extend perpendicularly (e.g., in the Z-direction) from opposite sides of the bottom wall 42.sub.1, 42.sub.1'. The second wall 42.sub.2, 42.sub.2' may include a first portion 42A, 42A' and/or the third wall 42.sub.3, 42.sub.3' may include a second portion 42B, 42B'. The first portion 42A, 42A' and/or the second portion 42B, 42B' may project laterally (e.g., in a Y-direction) toward a center of the track 40, 40'. The first portion 42A, 42A' and/or the second portion 42B, 42B' may be substantially planar. In embodiments, the first portion 42A, 42A' and/or the second portion 42B, 42B' may be disposed such that a gap 46, 46' may be provided between the first portion 42A, 42A' and the second portion 42B, 42B' (e.g., the first portion 42A, 42A' and the second portion 42B, 42B' may be offset in the Y-direction). The gap 46 may extend longitudinally along the track 40, and/or the gap 46 may be centered along the track 40.

(34) In embodiments, the inner track 44, 44' may be disposed at least partially in the outer track 42, 42'. The inner track 44, 44' may, for example and without limitation, be substantially U-shaped. The inner track 44, 44' may include a first wall 44.sub.1, 44.sub.1', a second wall 44.sub.2, 44.sub.2', and/or a third wall 44.sub.3, 44.sub.3'. The second wall 44.sub.2, 44.sub.2', may be shorter than the third wall 44.sub.3, 44.sub.3'. The second wall 44.sub.2, 44.sub.2' and/or the third wall 44.sub.3, 44.sub.3' may be at least partially bent and/or curved. The second wall 44.sub.2, 44.sub.2' and the third wall 44.sub.3, 44.sub.3' may extend perpendicularly (e.g., vertically) from the bottom wall 44.sub.1. The bottom wall 44.sub.1, 44.sub.1' of the inner track 44, 44' may be generally aligned with and/or adjacent to the bottom wall 42.sub.1, 42.sub.1' of the outer track 42, 42'. The second wall 44.sub.2, 44.sub.2' of the inner track 44, 44' may be generally aligned with and/or adjacent to the second wall 42.sub.2, 42.sub.2' of the outer track 42, 42'. The third wall 44.sub.3, 44.sub.3' of the inner track 44, 44' may be generally aligned with and/or adjacent to the third wall 42.sub.3, 42.sub.3' of the outer track 42, 42'.

(35) With embodiments, such as generally illustrated in FIGS. 2A and 2B, the outer track 42 of the first track 40 may include a first recess 48A and a third recess 48C. The outer track 42' of the second track 40' may include a second recess 48B and a fourth recess 48D. The recesses 48A, 48B, 48C, 48D may be disposed between a top of the second wall 42.sub.2, 42.sub.2' of the outer track 42, 42' and a top of the second wall 44.sub.2, 44.sub.2' of the inner track 44, 44'. The recesses 48A, 48B, 48C, 48D may extend partially into the second walls 42.sub.2, 42.sub.2' (e.g., in the Y-direction). The recesses 48A, 48B, 48C, 48D may include one or more of a variety of shapes, sizes, and/or configurations. For example and without limitation, the recesses 48A, 48B, 48C, 48D may

be substantially rectangular, circular, and/or curved.

(36) In embodiments, the first track **40** and/or the second track **40'** may include one or more bus bars **50A**, **50B**, **50C**, **50D** (e.g., electrical conductors). The first track **40** may include a first bus bar **50A** and/or a third bus bar **50C**. The second track **40'** may include a second bus bar **50B** and/or a fourth bus bar **50D**. The bus bars **50A**, **50B**, **50C**, **50D** may include one or more of a variety of shapes, sizes, and/or configurations. For example and without limitation, the bus bars **50A**, **50B**, **50C**, **50D** may be substantially U-shaped. The bus bars **50A**, **50B**, **50C**, **50D** may extend substantially longitudinally (e.g., in the X-direction). The bus bars **50A**, **50B**, **50C**, **50D** may be electrically conductive and/or include an electrically conductive material. The first bus bar **50A** may be disposed at least partially in the first recess **48A** of the first track **40**, the second bus bar **50B** may be disposed at least partially in the second recess **48B** of the second track **40'**, the third bus bar **50C** may be disposed at least partially in the third recess **48C** of the first track **40**, and/or the fourth bus bar **50D** may be disposed at least partially in the fourth recess **48D** of the second track **40'**. The bus bars **50A**, **50B**, **50C**, **50D** may be disposed at least partially between the outer tracks **42**, **42'** and the inner tracks **44**, **44'** (e.g., in the Z-direction). The bus bars **50A**, **50B**, **50C**, **50D** may extend along part of or along the entire length of the first track **40** and/or second track **40'**. The bus bars **50A**, **50B** may be electrically connected to a power source **26** (e.g., a vehicle battery) and may be configured to provide power from the power source **26** to the support member **30** at some or all points along the track **40**.

(37) With embodiments, the bus bars **50A**, **50B** may be configured for connection with a power source **26** and/or the first ECU **28A**. For example and without limitation, the first bus bar **50A** and/or the second bus bar **50B** may be configured for connection to the power source **26**. The first bus bar **50A** and/or the second bus bar **50B** may be configured to provide power to a support assembly **22** via the support member **30**. The first bus bar **50A** may connect to a first/positive terminal **26A** of the power source **26**, and/or the second bus bar **50B** may connect to a second/negative terminal **26B**, which may be connected to ground, of the power source **26**. The first bus bar **50A** and/or the second bus bar **50B** may supply power to the second ECU **28B** and/or to one or more electrical components **22C**, **22C.sub.1**, **22C.sub.2** (e.g., motors, heaters, fans, haptic devices, etc. as generally illustrated in FIGS. **1-2B**) of support assembly **22** that may provide one or more functions (e.g., support assembly movement, heating, cooling, massage, etc.).

(38) In embodiments, such as generally illustrated in FIGS. **2A** and **2B**, the support member **30** may include one or more conductors (e.g., conductors **36A**, **36B**, **36C**, **36D**). A first portion **32** of the support member **30** may include a first conductor **36A** and/or a third conductor **36C**. The second portion **34** of the support member **30** may include a second conductor **36B** and/or a fourth conductor **36D**. In a first orientation of the support assembly **22**, the first conductor **36A** may be configured for connection with the first bus bar **50A** and/or the second conductor **36B** may be configured for connection with the second bus bar **50B** (see, e.g., FIG. **2A**).

(39) With embodiments, in a second orientation of the support assembly **22**, the first conductor **36A** may be configured for connection with the second bus bar **50B** and/or the second conductor **36B** may be configured for connection with the first bus bar **50A** (see, e.g., FIG. **2B**). In embodiments, the conductors **36A**, **36B**, **36C**, **36D** may include one or more of a variety of shapes, sizes, and/or configurations. For example and without limitation, the conductors **36A**, **36B**, **36C**, **36D** may be oval-shaped, rectangular, curved, rounded, and/or oblong. The conductors **36A**, **36B**, **36C**, **36D** may be substantially planar.

(40) With embodiments, an electrical assembly **20** may include a first ECU **28A** and/or a second ECU **28B**. The first ECU **28A** and/or the second ECU **28B** may be configured to communicate with (e.g., receive information from, send information to, digitally communicate with, and/or sense a status/voltage of, etc.) the control circuit **52**, such as with the switch/relay assembly **60**, the diode assembly **80**, and the switch assembly **90**. The first ECU **28A** and/or the second ECU **28B** may be configured to sense the status of the control circuit **52** (e.g., such as voltage). The first ECU **28A**

may be connected to the track assembly **38**. The second ECU **28B** may be connected to the support member **30**. The second ECU **28B** may, for example and without limitation, be configured to control one or more functions/electrical components **22C** of the support assembly **22**. The control circuit **52** may be connected between (e.g., electrically) the bus bars **50A**, **50B** and the second ECU **28B**. The first ECU **28A** may be configured to receive information about the orientation of the support member **30** (and a seat **30A** that may be connected thereto), such as via the second ECU **28B** and/or the control circuit **52**. For example and without limitation, the first ECU **28A** may be configured to receive information from the second ECU **28B** and/or the control circuit **52** indicating whether the support assembly **22** is forward facing or rearward facing.

(41) In embodiments, the control circuit **52** may include a switch/relay assembly **60**. The relay assembly **60** may be configured to connect the appropriate support assembly terminals **22A**, **22B** to the power source **26**. For example and without limitation, the relay assembly **60** may be configured to connect the correct terminals **26A**, **26B** of the power source **26** to the appropriate support assembly terminals **22A**, **22B** (e.g., such that the first terminal **22A** of the support assembly **22** is connected to the first terminal **26A** of the power source **26** and the second terminal **22B** is connected to the second terminal **26B** of the power source **26**, regardless of the orientation of the support assembly **22**). The relay assembly **60** may include one or more relays (e.g., relays **62**, **64**) and/or one or more diodes (e.g., diode **74**). The one or more relays may, for example and without limitation, include one or more electromechanical relays and/or one or more solid state relays. Upon connecting the support member **30** to the track **40**, the relay assembly **60** may automatically connect the positive terminal **26A** of the power source **26** to a positive terminal **22A** of the support assembly **22**. Additionally or alternatively, the relay assembly **60** may automatically connect the negative (e.g., ground) terminal **26B** of the power source **26** to the negative terminal **22B** of the support assembly **22**. The relay assembly **60** may be disposed at least partially in the support member **30** and/or in the seat **30A**.

(42) With embodiments, such as generally shown in FIGS. **3A**, **3B**, and **3C**, the control circuit **52** (e.g., the relay assembly **60**) may be configured to automatically connect the power source **26** to the correct terminals **22A**, **22B** of the support assembly **22** regardless of the orientation of the support assembly **22**. The relay assembly **60** may include a first relay **62**, a second relay **64**, a first diode **66**, and/or a second diode **68**. The first relay **62** and/or the second relay **64** may include a first contact **62.sub.1**, **64.sub.1**, a second contact **62.sub.2**, **64.sub.2**, a third contact **62.sub.3**, **64.sub.3**, a fourth contact **62.sub.4**, **64.sub.4**, and/or a fifth contact **62.sub.5**, **64.sub.5**. The relays **62**, **64** may be configured to selectively electrically connect the first contacts **62.sub.1**, **64.sub.1** with the second contacts **62.sub.2**, **64.sub.2** or the third contacts **62.sub.3**, **64.sub.3**.

(43) In embodiments, the first relay **62** may be connected to the first diode **66** and/or the second relay **64** may be connected to the second diode **68**. The first relay **62** (e.g., the first contact **62.sub.1**) may be connected to the positive terminal **22A** of the support assembly **22**, and/or the second relay **64** (e.g., the first contact **64.sub.1**) may be connected to the negative terminal **22B** (e.g., ground) of the support assembly **22**. The first relay **62** and/or the second relay **64** may include a first coil **70** and/or a second coil **72**, respectively. The coils **70**, **72** may be connected between the fourth contacts **62.sub.4**, **64.sub.4** and the fifth contacts **62.sub.5**, **64.sub.5** of the first relay **62** and the second relay **64**, respectively. The first diode **66** may be connected to the fifth contact **62.sub.5** of the first relay **62**. The first diode **66** may permit current flow into the fifth contact **62.sub.5** and/or may restrict current flow out of the fifth contact **62.sub.5**. The second diode **68** may be connected to the fifth contact **64.sub.5** of the second relay **64**. The second diode **68** may permit current flow into the fifth contact **64.sub.5** and/or may restrict current flow out of the fifth contact **64.sub.5**. The second contact **62.sub.2** of the first relay **62** may be connected to the fourth contact **62.sub.4** of the first relay **62**, the second contact **64.sub.2** of the second relay **64**, the second diode **68**, and/or the second conductor **36B**. The third contact **62.sub.3** of the first relay **62** may be connected to the first diode **66**, the fourth contact **64.sub.4** of the second relay **64**, the third contact

64.sub.3 of the second relay 64, the first conductor 36A, and/or the second ECU 28B (e.g., to output support assembly position information).

(44) With embodiments, such as generally shown in FIG. 3A, the relay assembly 60 may include a first state (e.g., an initial state). When the relay assembly 60 is in the first state, the support assembly 22 may not be connected to the track assembly 38, and/or the support assembly 22 may not be connected to the power source 26. In the first state of the relay assembly 60, the first contact 62.sub.1 of the first relay 62 may be connected to the second contact 62.sub.2, and/or the first contact 64.sub.1 of the second relay 64 may be connected to the second contact 64.sub.2.

Additionally or alternatively, in the first state of the relay assembly 60, the first conductor 36A and/or the second conductor 36B may not be connected to the first bus bar 50A and/or the second bus bar 50B.

(45) In embodiments, such as generally illustrated in FIG. 3B, the relay assembly 60 may include a second state which may correspond to the support assembly 22 being disposed in a first/forward-facing orientation and connected to the track assembly 38. When the relay assembly 60 is in the second state, the support assembly 22 may be connected to the track 40 and/or the support assembly 22 may be connected to the power source 26, such as via the first conductor 36A that may be connected to the first bus bar 50A (which may be connected to the positive terminal 26A) and/or via the second conductor 36B that may be connected to the second bus bar 50B (which may be connected to the negative terminal 26B).

(46) With embodiments, connecting the positive terminal 26A of the power source 26 to the first conductor 36A may cause the first coil 70 to trigger (e.g., energize), which may connect the first contact 62.sub.1 of the first relay 62 to the third contact 62.sub.3 instead of the second contact 62.sub.2. Current may flow from the positive terminal 26A to the first bus bar 50A, to the first conductor 36A, through the first relay 62, and/or to the positive terminal 22A of the support assembly 22. The second coil 72 may not be energized, and/or the second diode 68 may prevent the second coil 72 from energizing when the first coil 70 is energized. In the second state, the first contact 64.sub.1 and second contact 64.sub.2 of the second relay 64 may remain connected. For example and without limitation, current may flow from the negative terminal 22B to the first contact 64.sub.1, to the second contact 64.sub.2, to the second conductor 36B, to the second bus bar 50B, and/or to the negative terminal 26B of the power source 26.

(47) In embodiments, such as generally illustrated in FIG. 3C, the relay assembly 60 may include a third state that may correspond to the support assembly 22 being disposed in a rearward-facing orientation. When the relay assembly 60 is in the third state, the support assembly 22 may be connected to the track 40 and/or the support assembly 22 may be connected to the power source 26. When the relay assembly 60 is in the third state, the first bus bar 50A may be connected to the second conductor 36B, which may connect the positive terminal 26A of the power source 26 to the second conductor 36B. Additionally or alternatively, in the third state of the relay assembly 60, the second bus bar 50B may be connected to the first conductor 36A, which may connect the negative terminal 26B of the power source 26 to the first conductor 36A.

(48) With embodiments, connecting the positive terminal 26A of the power source 26 to the second conductor 36B may cause the second coil 72 to trigger (e.g., energize), which may connect the first contact 64.sub.1 of the second relay 64 to the third contact 64.sub.3 instead of the second contact 64.sub.2. Current may flow from positive terminal 26A to the first bus bar 50A, to the second conductor 36B, through the first relay 62, and/or to the positive terminal 22A of the support assembly 22. The first coil 70 may not be energized, and/or the first diode 66 may prevent the first coil 70 from energizing when the second coil 72 is energized. The second bus bar 50B may be connected to the third contact 64.sub.3 of the second relay 64 and/or may be connected to the first contact 64.sub.1 of the second relay 64 such as to connect to the negative terminal 22B of the support assembly 22.

(49) In embodiments, such as generally shown in FIGS. 4A and 4B, a control circuit 52 (e.g., relay

assembly 60) may be configured to automatically connect the power source 26 to the correct terminals 22A, 22B of the support assembly 22 regardless of orientation. The relay assembly 60 may include a first relay 62, a second relay 64, and/or a diode 74. The first relay 62 and/or the second relay 64 may be connected to the diode 74. The first relay 62 (e.g., the first contact 62.sub.1) may be connected to the positive terminal 22A of the support assembly 22, and/or the second contact 64.sub.2 of the second relay 64 may be connected to the negative terminal 22B of the support assembly 22. The diode 74 may be connected to the fifth contact 62.sub.5 of the first relay 62 and the fifth contact 64.sub.5 of the second relay 64. The diode 74 may permit current flow from the second conductor 36B to the fifth contacts 62.sub.5, 64.sub.5 of the relays 62, 64 (and from the fifth contacts 62.sub.5, 64.sub.5 to the fourth contacts 62.sub.4, 64.sub.4), and/or may restrict current flow from the fifth contacts 62.sub.5, 64.sub.5 to the second conductor 36B (e.g., to prevent energizing the coils 70, 72 in the first orientation). The second contact 62.sub.2 of the first relay 62 may be connected to the first conductor 36A, the fourth contact 62.sub.4 of the first relay 62, the fourth contact 64.sub.4 of the second relay 64, the third contact 64.sub.3 of the second relay 64, and/or the second ECU 28B (e.g., to output seat position or other information). The third contact 62.sub.3 of the first relay 62 may be connected to the second conductor 36B, the diode 74, and/or the second contact 64.sub.2 of the second relay 64.

(50) With embodiments, such as generally illustrated in FIG. 4A, the relay assembly 60 may include a first state that may correspond to the support assembly 22 being disposed in a first/forward-facing orientation. When the relay assembly 60 is in the first state, the support assembly 22 may be connected to the track 40 and/or support assembly 22 may be connected to the power source 26. When the relay assembly 60 is in the first state, the first bus bar 50A may be connected to the first conductor 36A, which may connect the first conductor 36A with the positive terminal 26A of the power source 26. Additionally or alternatively, in the first state of the relay assembly 60, the second bus bar 50B may be connected to the second conductor 36B, which may connect the second conductor 36B with the negative terminal 26B of the power source 26.

(51) In embodiments, connecting the positive terminal 26A of the power source 26 to the first conductor 36A may not cause the first coil 70 and/or the second coil 72 to trigger (e.g., energize). The first contacts 62.sub.1, 64.sub.1 of the relays 62, 64 may remain connected to the second contacts 62.sub.2, 64.sub.2. Current may flow from the positive terminal 26A of the power source 26 to the first bus bar 50A, to the first conductor 36A, through the first relay 62, and/or to the positive terminal 22A of the support assembly 22. The first coil 70 and/or the second coil 72 may not be energized, and/or the diode 74 may prevent the first coil 70 and/or the second coil 72 from energizing when the support assembly 22 is in the forward-facing orientation. The second bus bar 50B may be connected via the second conductor 36B to the second contact 64.sub.2 of the second relay 64 and/or may be connected to the first contact 64.sub.1 of the second relay 64, such as to connect the negative terminal 26B of the power source 26 to the negative terminal 22B of the support assembly 22.

(52) With embodiments, such as generally illustrated in FIG. 4B, the relay assembly 60 may include a second state that may correspond to the support assembly 22 being disposed in a second/rearward facing orientation. When the relay assembly 60 is in the second state, the support assembly 22 may be connected to the track 40 and/or the support assembly 22 may be connected to the power source 26. When the relay assembly 60 is in the second state, the first bus bar 50A may be connected to the second conductor 36B, which may connect the second conductor 36B with the positive terminal 26A of the power source 26. Additionally or alternatively, in the second state of the relay assembly 60, the second bus bar 50B may be connected to the first conductor 36A, which may connect the first conductor 36A with the negative terminal 26B of the power source 26.

(53) In embodiments, connecting the positive terminal 26A of the power source 26 to the second conductor 36B may cause the first coil 70 and/or the second coil 72 to trigger (e.g., trip/energize) which may connect the first contacts 62.sub.1, 64.sub.1 of the relays 62, 64 to the third contacts

62.sub.3, 64.sub.3 instead of the second contacts **62.sub.2, 64.sub.2**. Current may flow from the positive terminal **26A** to the first bus bar **50A**, to the second conductor **36B**, through the first relay **62**, and/or to the positive terminal **22A** of the support assembly **22**. The diode **74** may prevent current from flowing through the second relay **64** to the negative terminal **22B** of the support assembly **22**. The second bus bar **50B** (e.g., ground) may be connected to the third contact **64.sub.3** of the second relay **64**, which may be connected to the first contact **64.sub.1** of the second relay **64** (which may be connected to the negative terminal **22B** of the support assembly **22**).

(54) With embodiments, such as generally illustrated in FIGS. **5A** and **5B**, a control circuit **52** (e.g., the relay assembly **60**) may include a first relay **62**, a second relay **64**, a third relay **76**, a first diode **74**, and/or a second diode **78** (e.g., a pulse diode). The third relay **76** may include a first contact **76.sub.1**, a second contact **76.sub.2**, a third contact **76.sub.3**, a fourth contact **76.sub.4**, and/or a fifth contact **76.sub.5**. The third relay **76** may include a third coil **76A** that may be connected between the fourth contact **76.sub.4** and the fifth contact **76.sub.5**. The first contact **76.sub.1** may be connected to the second contact **76.sub.2**, which may be configured as an open contact, when the third coil **76A** is not energized, and/or the first contact **76.sub.1** may be connected to the third contact **76.sub.3** when the third coil **76A** is energized. The first contact **76.sub.1** of the third relay **76** may be connected to the positive terminal **22A** of the support assembly **22**. The fourth contact **76.sub.4** of the third relay **76** may be connected to the first contact **64.sub.1** of the second relay **64** and/or the negative terminal **22B** (e.g., ground) of the support assembly **22**. The pulse diode **78** may be connected to the third contact **76.sub.3** of the third relay **76**, the fifth contact **76.sub.5** of the third relay **76**, and/or the first contact **62.sub.1** of the first relay **62**. The pulse diode **78** may be configured to permit current flow into the fifth contact **76.sub.5** of the third relay **76** and/or may restrict or prevent current flow out from the fifth contact **76.sub.5** (e.g., to prevent energizing the coils **70, 72** in the first orientation). The first contact **62.sub.1** of the first relay **62** may be connected to the third contact **76.sub.3** of the third relay **76**.

(55) In embodiments, the third relay **76** and/or the pulse diode **78** of the relay assembly **60** may isolate the first relay **62** and/or the second relay **64** during switching (e.g., coil energizing). Switching the polarity of the contacts at the first conductor **36A** and/or second conductor **36B** may result in a reverse battery pulse. The pulse diode **78** and/or third relay **76** switching delay may limit the reverse battery pulse from affecting the support assembly **22** (e.g., internal circuitry of the support assembly **22**, the second ECU **28B**, and/or electrical components **22C** that may be connected to the support assembly **22**).

(56) In embodiments, such as generally illustrated in FIG. **5A**, the relay assembly **60** may include a first state that may correspond to the support assembly **22** being disposed in a first/forward-facing orientation. When the relay assembly **60** is in the first state, the support assembly **22** may be connected to the track **40** and/or the support assembly **22** may be connected to the power source **26**. When the relay assembly **60** is in the first state, the first bus bar **50A** may be connected to the first conductor **36A**, which may connect the first conductor **36A** with the positive terminal **26A** of the power source **26**. Additionally or alternatively, in the first state of the relay assembly **60**, the second bus bar **50B** may be connected to the second conductor **36B**, which may connect the second conductor **36B** with the negative terminal **26B** of the power source **26**.

(57) With embodiments, connecting the positive terminal **26A** of the power source **26** to the first conductor **36A** may not cause the first coil **70** and/or the second coil **72** to trigger (e.g., energize). The first contacts **62.sub.1, 64.sub.1** of the relays **62, 64** may remain connected to the second contacts **62.sub.2, 64.sub.2**. Connecting the positive terminal **22A** to the first conductor **36A** may cause the third coil **76A** to energize. For example and without limitation, current may flow from the positive terminal **26A**, to the first bus bar **50A**, to the first conductor **36A**, to the first contact **62.sub.1** of the first relay **62**, to the pulse diode **78**, to the fifth contact **76.sub.5** of the third relay **76**, and to the third coil **76A**, which may energize the third coil **76A**. Energizing the third coil **76A** may cause the first contact **76.sub.1** of the third relay **76** to disconnect from the second contact

76.sub.2 and connect to the third contact **76.sub.3**, which may connect the positive terminal **26A** of the power source **26** to the positive terminal **22A** of the support assembly **22**.

(58) In embodiments, such as generally illustrated in FIG. **5B**, the relay assembly **60** may include a second state that may correspond to the support assembly **22** being disposed in a second/rearward-facing orientation. When the relay assembly **60** is in the second state, the support assembly **22** may be connected to the track assembly **38** and/or support assembly **22** may be connected to the power source **26**. When the relay assembly **60** is in the second state, the first bus bar **50A** may be connected to the second conductor **36B**, which may connect the second conductor **36B** with the positive terminal **26A** of the power source **26**. Additionally or alternatively, in the second state of the relay assembly **60**, the second bus bar **50B** may be connected to the first conductor **36A**, which may connect the first conductor **36A** with the negative terminal **26B** of the power source **26**.

(59) With embodiments, connecting the positive terminal **26A** of the power source **26** to the second conductor **36B** may cause the first coil **70** and/or the second coil **72** to trigger (e.g., energize). The first contacts **62.sub.1**, **64.sub.1** of the first relay **62** and the second relay **64** may disconnect from the second contacts **62.sub.2**, **64.sub.2** and/or may connect to the third contacts **62.sub.3**, **64.sub.3**. Current may flow from the positive terminal **26A** of the power source **26** to the first bus bar **50A**, to the second conductor **36B**, to the third contact **62.sub.3** of the first relay **62**, to the first contact **62.sub.1** of the first relay **62**, to the third contact **76.sub.3** of the third relay **76**, and/or to the pulse diode **78**, which may energize the third coil **76A**. Energizing the third coil **76A** may cause the first contact **76.sub.1** of the third relay **76** to disconnect from the second contact **76.sub.2** and connect to the third contact **76.sub.3**, which may connect the positive terminal **26A** of the power source **26** to the positive terminal **22A** of the support assembly **22** via the second conductor **36B**.

(60) In embodiments, such as generally illustrated in FIGS. **6A** and **6B**, a control circuit **52** may include a diode assembly **80**. The diode assembly **80** may include diodes (e.g., diodes **82.sub.1**, **82.sub.2**, **82.sub.3**, **82.sub.4**) and may or may not include electromechanical components such as relays and/or switches. The diode assembly **80** may be configured to connect the correct terminals **26A**, **26B** of the power source **26** to the appropriate support assembly terminals **22A**, **22B** (e.g., such that the first terminal **22A** of the support assembly **22** is connected to the first terminal **26A** of the power source **26** and the second terminal **22B** is connected to the second terminal **26B** of the power source **26**, regardless of the orientation of the support assembly **22**). The diode assembly **80** may include a first diode **82.sub.1**, a second diode **82.sub.2**, a third diode **82.sub.3**, and/or a fourth diode **82.sub.4**. The first diode **82.sub.1**, the second diode **82.sub.2**, the third diode **82.sub.3**, and the fourth diode **82.sub.4** may be connected as a bridge circuit. The first conductor **36A** may be connected between the first diode **82.sub.1** and the second diode **82.sub.2**. The second conductor **36B** may be connected between the third diode **82.sub.3** and the fourth diode **82.sub.4**. A diode assembly **80** may include and/or be connected to one or more other passive electrical components (e.g., additional diodes or other components).

(61) In embodiments, such as generally illustrated in FIG. **6A**, if the support assembly **22** is in a first/forward orientation, the first conductor **36A** may be connected to the first bus bar **50A** and the positive terminal **26A** of the power source **26** and/or the second conductor **36B** may be connected to the second bus bar **50B** and the negative terminal **26B** of the power source **26**. Current may flow from the positive terminal **26A** to the first bus bar **50A**, to the first conductor **36A**, through the first diode **82.sub.1**, and to the first terminal **22A** of the support assembly **22**. In the first orientation, the second diode **82.sub.2** and/or the third diode **82.sub.3** may block current from flowing from the positive terminal **26A** to the second terminal **22B** of the support assembly **22**. In the first orientation, current may flow from the second terminal **22B** of the support assembly **22** through the fourth diode **82.sub.4** to the second conductor **36B**, the second bus bar **50B**, and/or the negative terminal **26B** of the power source **26**.

(62) In embodiments, such as generally illustrated in FIG. **6B**, if the support assembly **22** is in a second/rearward orientation, the first conductor **36A** may be connected to the second bus bar **50B**

and the negative terminal 26B of the power source 26, and/or the second conductor 36B may be connected to the first bus bar 50A and the positive terminal 26A of the power source 26. Current may flow from the positive terminal 26A to the first bus bar 50A to the second conductor 36B, through the third diode 82.sub.3 and to the first terminal 22A of the support assembly 22. In the second orientation, the first diode 82.sub.1 and/or the fourth diode 82.sub.4 may block current from flowing from the positive terminal 26A to the second terminal 22B of the support assembly 22. In the second orientation, current may flow from the second terminal 22B of the support assembly 22 through the second diode 82.sub.2 to the first conductor 36A, the second bus bar 50B, and/or the negative terminal 26B of the power source 26.

(63) With embodiments, such as generally illustrated in FIGS. 7A and 7B, a control circuit 52 may include a switch assembly 90. The switch assembly 90 may be configured to connect the power source 26 to the support assembly 22 with the correct polarity regardless of the orientation of the support assembly 22. For example and without limitation, the switch assembly 90 may be configured to connect the correct terminals 26A, 26B of the power source 26 to the appropriate support assembly terminals 22A, 22B (e.g., such that the first terminal 22A of the support assembly 22 is connected to the first terminal 26A of the power source 26 and the second terminal 22B is connected to the second terminal 26B of the power source 26, regardless of the orientation of the support assembly 22).

(64) With embodiments, switches of the switch assembly 90 may include one or more of a variety of configurations. The switch assembly 90 may include switches (e.g., the switch assembly 90 may or may not include electromechanical components such as electromechanical relays). For example and without limitation, the switch assembly 90 may include a first switch 92.sub.1, a second switch 92.sub.2, a third switch 92.sub.3, and/or a fourth switch 92.sub.4 that may include silicon-based switches, transistors, and/or metal-oxide field effect transistors (MOSFETs), among other configurations. The first switch 92.sub.1 may be connected to a first driver 94.sub.1, the second switch 92.sub.2 may be connected to a second driver 94.sub.2, the third switch 92.sub.3 may be connected to a third driver 94.sub.3, and/or the fourth switch 92.sub.4 may be connected to a fourth driver 94.sub.4. The first switch 92.sub.1, the second switch 92.sub.2, the third switch 92.sub.3, and the fourth switch 92.sub.4 may be connected as a bridge circuit. The drivers 94.sub.1, 94.sub.2, 94.sub.3, 94.sub.4 may be configured to activate the switches 92.sub.1, 92.sub.2, 92.sub.3, 92.sub.4, respectively. The drivers 94.sub.1, 94.sub.2, 94.sub.3, 94.sub.4 may not carry power, but may set the gate voltage of the switches 92.sub.1, 92.sub.2, 92.sub.3, 92.sub.4 such that the switches 92.sub.1, 92.sub.2, 92.sub.3, 92.sub.4 may selectively open. The first conductor 36A may be connected between the first switch 92.sub.1 and the second switch 92.sub.2. The second conductor 36B may be connected between the third switch 92.sub.3 and the fourth switch 92.sub.4. The switch assembly 90 may include and/or be connected to one or more other passive electrical components (e.g., additional switches, one or more diodes, etc.).

(65) In embodiments, such as generally illustrated in FIG. 7A, if the support assembly 22 is in a first/forward orientation, the first conductor 36A may be connected to the first bus bar 50A and the positive terminal 26A of the power source 26 and/or the second conductor 36B may be connected to the second bus bar 50B and the negative terminal 26B of the power source 26. Current may flow from the positive terminal 26A to the first bus bar 50A, to the first conductor 36A, and to the first switch 92.sub.1 and the first driver 94.sub.1. The first driver 94.sub.1 may activate the first switch 92.sub.1 to allow current to flow to the first terminal 22A of the support assembly 22. In the first orientation, the second switch 92.sub.2 and/or the third switch 92.sub.3 may not be activated and may block current from flowing from the positive terminal 26A to the second terminal 22B of the support assembly 22. In the first orientation, current may flow from the second terminal 22B of the support assembly 22 to the fourth switch 92.sub.4 and the fourth driver 94.sub.4. The fourth driver 94.sub.4 may activate the fourth switch 92.sub.4 to allow current to flow from the fourth switch 92.sub.4 to the negative terminal 26B of the power source 26.

(66) In embodiments, such as generally illustrated in FIG. 7B, if the support assembly 22 is in a second/rearward orientation, the first conductor 36A may be connected to the second bus bar 50B and the negative terminal 26B of the power source 26, and/or the second conductor 36B may be connected to the first bus bar 50A and the positive terminal 26A of the power source 26. Current may flow from the positive terminal 26A to the first bus bar 50A, to the second conductor 36B, and to the third switch 92.sub.3 and the third driver 94.sub.3. The third driver 94.sub.3 may activate the third switch 92.sub.3 to allow current to flow to the first terminal 22A of the support assembly 22. In the second orientation, the first switch 92.sub.1 and/or the fourth switch 92.sub.4 may not be activated and may block current from flowing from the positive terminal 26A to the second terminal 22B of the support assembly 22. In the second orientation, current may flow from the second terminal 22B of the support assembly 22 to the second switch 92.sub.2 and the second driver 94.sub.2. The second driver 94.sub.2 may activate the second switch 92.sub.2 to allow current to flow to the first conductor 36A, the second bus bar 50B, and/or the negative terminal 26B of the power source 26.

(67) With embodiments, switches and drivers of a switch assembly 90 (e.g., switches 92.sub.1, 92.sub.2, 92.sub.3, 92.sub.4 and drivers 94.sub.1, 94.sub.2, 94.sub.3, 94.sub.4) may be configured for automatic activation (e.g., independent of any separate controllers, such as the ECUs 28A, 28B). If the correct polarity is provided to a switch and a driver, the driver may automatically activate the switch. If the reverse polarity is provided to the switch and the driver, the driver may not activate the switch. The switches 92.sub.1, 92.sub.2, 92.sub.3, 92.sub.4 may, for example and without limitation, be connected in a bridge configuration.

(68) In embodiments, a control circuit 52, a diode assembly 80, and/or a switch assembly 90 may include at least four electrical components (e.g., non-electromechanical components) configured to connect the correct terminals 26A, 26B of the power source 26 to the appropriate support assembly terminals 22A, 22B regardless of the orientation of the support assembly 22.

(69) With embodiments, such as generally illustrated in FIGS. 7C and 7D, the first driver 94.sub.1, the second driver 94.sub.2, the third driver 94.sub.3, and the fourth driver 94.sub.4 may be combined into any number of drivers to control the switches 92.sub.1, 92.sub.2, 92.sub.3, 92.sub.4 (see, e.g., FIG. 7C for two drivers and see FIG. 7D for one driver). For example and without limitation, the first driver 94.sub.1, the second driver 94.sub.2, the third driver 94.sub.3, and the fourth driver 94.sub.4 may be combined into a first driver 96.sub.1 and a second driver 96.sub.2 (see, e.g., FIG. 7C). The first driver 96.sub.1 may be connected to the first switch 92.sub.1 and the second switch 92.sub.2. The second driver 96.sub.2 may be connected to the third switch 92.sub.3 and the fourth switch 92.sub.4. The first driver 96.sub.1 may be configured to control/activate the first switch 92.sub.1 and the second switch 92.sub.2. The second driver 96.sub.2 may be configured to control/activate the third switch 92.sub.3 and the fourth switch 92.sub.4. In embodiments, the control circuit 52, may include a single driver 98 that may be configured to control/activate the first switch 92.sub.1, the second switch 92.sub.2, the third switch 92.sub.3, and the fourth switch 92.sub.4 (see, e.g., FIG. 7D). The driver 98 may be connected to each of the switches 92.sub.1, 92.sub.2, 92.sub.3, 92.sub.4.

(70) Embodiments of control circuits 52, such as the five embodiments illustrated in FIGS. 3A-3C, 4A and 4B, 5A and 5B, 6A and 6B, and 7A-7D, may include various advantages and/or potential drawbacks. The embodiment of FIGS. 3A-3C may, for example and without limitation, include a compact configuration, may involve medium cost, and may involve a relatively low voltage drop, but may experience reduced relay life cycle (e.g., each maneuver of the support assembly 22 may actuate a relay) and/or may involve increased noise from the relays 62, 64. The embodiment of FIGS. 4A and 4B may, for example and without limitation, involve low cost, a relatively low voltage drop, a relative long relay life cycle/minimal relay noise (e.g., relays may only be actuated when a support assembly 22 is disposed in a second orientation), but may experience a reverse pulse from the power source 26 during an initial connection of the support member 30 with the

track assembly **38**. The embodiment of FIGS. **5A** and **5B** may, for example and without limitation, not involve a reverse pulse from the power source **26**, but may involve a higher cost, a higher voltage drop, and/or may experience a shorter life cycle for the third relay **76** which may actuate for each maneuver of the support assembly **22**. The embodiment of FIGS. **6A** and **6B** may, for example and without limitation, involve high cost, a relatively high voltage drop, a relatively long diode life (e.g., longer than the expected life of a vehicle), a no-noise and compact control circuit **52**. The embodiment of FIGS. **7A-7D** may, for example and without limitation, involve a high cost, a relatively low voltage drop, minimum power waste, a longer circuit life (e.g., longer than the expected life of a vehicle), low or substantially no noise, and/or a compact control circuit **52**.

(71) In embodiments, an electrical assembly **20** may be configured to avoid a reverse polarity conduction, provide power to the support assembly **22** in the second/rearward-facing configuration, and/or provide digital monitoring of the position of the support assembly **22**.

(72) With embodiments, a control circuit **52** may operate automatically, such as independently of the ECUs **28A**, **28B**. For example and without limitation, a control circuit **52** (e.g., a relay assembly) **60** may switch between states (e.g., a first state, a second state, and/or a third state) without being controlled by an ECU **28A**, **28B**. One or both of the ECUs **28A**, **28B** may be connected to the control circuit **52** and the connection may be a passive/monitoring connection. A control circuit **52** may be configured as a passive assembly and may not involve a capacitor or internal energy storage.

(73) With embodiments, such as generally illustrated in FIGS. **8-19**, it may be desirable to limit or remove involvement of an ECU (e.g., ECU **28**) when providing/ensuring a proper electrical connection (e.g., with the correct polarity) between a track and a support member, and/or between a power source **190** and a load **160**. For example and without limitation, limiting or removing involvement of an ECU may reduce complexity and/or power consumption.

(74) In embodiments, such as generally illustrated in FIGS. **8-10**, an electrical track assembly **120** may include a pair of tracks **122** and one or more support members, such as a first support member **124** and/or a second support member **124'**. The one or more support members **124**, **124'** may be configured for selective connection with, removal from, and/or movement along the pair of tracks **122**, such as in one or more configurations. For example and without limitation, the one or more support members **124** may be configured for selective connection with, removal from, and/or movement along the pair of tracks **122** in a first configuration (e.g., facing a first direction, such as forward—see first support member **124** in FIGS. **8** and **9**) and a second configurations (e.g., facing a second direction, such as rearward—see second support member **124'** in FIG. **8** and first support member **124** in FIG. **10**). A support member **124**, **124'** may, for example, be connected to the pair of tracks **122** in the first configuration, move along the tracks **122** while in the first configuration, be removed from the pair of tracks **122** while in the first configuration, and then be connected with the tracks **122** in the second configuration.

(75) With embodiments, the pair of tracks/rails **122** may include a first track/rail **130** and a second track/rail **132** that may be disposed in parallel with each other, such as on a mounting surface **140** (e.g., a floor of a vehicle **100**), and may be offset from each other, such as in a Y-direction (e.g., a lateral direction). The pair of tracks **122** may be separately connected (e.g., fixed) to the mounting surface **140**. The mounting surface **140** may include a first side **142** (e.g., a left side), a second side **144** (e.g., a right side), a first end **146** (e.g., a front end), and/or a second end **148** (e.g., a rear end). A support member **124**, **124'** may be configured for connection with both of the first track **130** and the second track **132** when connected in the first configuration or the second configuration. For example and without limitation, a support member **124**, **124'** may include a first connection portion **150** that may be configured for connection with the first track **130** in the first configuration and the second track **132** in the second configuration, and/or may include a second connection portion **152** that may be configured for connection with the second track **132** in the first configuration and the first track **130** in the second configuration. The tracks **130**, **132** may include a generally U-shaped

configuration that may open upward (e.g., in a Z-direction) and the connection portions **150, 152** may extend into and/or engage the tracks **130, 132**.

(76) With embodiments, such as generally illustrated in FIGS. **8-10**, a support member **124, 124'** may include an electrical load **160**. An electrical load **160** may, for example and without limitation, include one or more of an electronic controller, an electric motor, a heater, a fan, fluid bladders, and/or pumps, among others. The first track **130** may include a first conductor **170** and/or the second track **132** may include a second conductor **172** that may be configured for providing electrical power to the electrical load **160** when the support member **124, 124'** is connected with the tracks **130, 132**. The first conductor **170** and/or the second conductor **172** may include electrically conductive material that may extend along some, most, or all of the respective track **130, 132**, and/or may be configured as bus bars. The conductors **170, 172** may be connected (e.g., attached) to and/or disposed at least partially in a wall (e.g., a substantially vertical wall) of a respective track **130, 132**.

(77) In embodiments, such as generally illustrated in FIGS. **9** and **10**, a support member **124, 124'** may include one or more electrical contacts, such as a first support member contact **174** and/or a second support member contact **176** that may be configured to provide an electrical connection between the support member **124, 124'** and the conductors **170, 172**. The first support member contact **174** may be configured to move (e.g., rotate) into and out of contact with the first conductor **170** and/or the second conductor **172**. The second support member contact **176** may be configured to move (e.g., rotate) into and out of contact with the first conductor **170** and/or the second conductor **172**. Movement of the support member contacts **174, 176** may, for example, be conducted manually (e.g., by a user), mechanically (e.g., via one or more levers, sliders, linkages, etc. upon connection with the tracks **130, 132**), and/or electromechanically (e.g., via an electric motor).

(78) With embodiments, a support member **124, 124'** may include a switch assembly **180** (e.g., a manual switch assembly) that may selectively electrically connect the electrical load **160** with the conductors **170, 172**, such as via the support member contacts **174, 176**. The electrical load **160** may include a first load contact **162** and/or a second load contact **164**. The first load contact **162** may, for example, include a positive contact. The second load contact **164** may, for example, include a negative contact. The first conductor **170** may be a positive conductor that may be connected to a positive power source contact **192** of a power source **190**. The second conductor **172** may be a negative conductor that may be connected to a negative power source contact **194** of the power source **190**. The power source **190** may, for example and without limitation, include a battery and/or an electrical generator, among others.

(79) In embodiments, a switch assembly **180** may be configured to connect the first load contact **162** with the first conductor **170** and connect the second load contact **164** with the second conductor **172**, such as regardless of the configuration of the support member **124, 124'**. The electrical track assembly **120** and/or the switch assembly **180** may be configured to prevent current flow between the first load contact **162** and the second conductor **172**, and/or prevent current flow between the second load contact **164** and the first conductor **170** (e.g., to prevent a reversed polarity connection).

(80) With embodiments, such as generally illustrated in FIGS. **12-14**, a switch assembly **180** may be configured to move between a first position (see, e.g., switch assembly **180** of support member **124** in FIG. **12**), a second position (see, e.g., switch assembly **180** of support member **124'** in FIG. **12**), and/or a third position (see, e.g. switch assemblies **180** in FIG. **14**). The first position may correspond to the first support member contact **174** being disposed proximate a first side **124A** of the support member **124, 124'** and the second support member contact **176** being disposed proximate a second side **124B** of the support member **124, 124'**. The second position may correspond to the first support member contact **174** being disposed proximate the second side **124B** of the support member **124, 124'** and the second support member contact **176** being disposed at the

first side **24A** of the support member **124, 124'**. The third position may correspond to a neutral position in which the first support member contact **174** and the second support member contact **176** may be disposed proximate a middle of the support member **124, 124'**. The first support member contact **174** and the second support member contact **176** may move with the switch assembly **180** such as to the first position, the second position, and/or the third position (e.g., may be manually operated via actuation of the switch assembly **180**).

(81) In embodiments, a switch assembly **180** may be configured to be moved (e.g., by a user) from a disengaged position (e.g., the third/neutral position) in which the support member **124, 124'** is not electrically connected with first conductor **170** and the second conductor **172**, to an engaged position (e.g., the first position and/or the second position) in which the support member **124, 124'** is electrically connected with first conductor **170** and the second conductor **172** (and vice versa). For example and without limitation, contact between the support member contacts **174, 176**, and the conductors **170, 172** may (e.g., immediately) provide an electrical connection for and/or power to the electrical load **160**, if properly switched/actuated.

(82) In embodiments, such as generally illustrated in FIG. **12**, if a support member **124** is connected with the tracks in a first configuration, the first side **124A** of the support member **124**, may be disposed proximate the first track **130**, and the second side **124B** of the support member **124** may be disposed proximate the second track **132**. In the first configuration, the switch assembly **180** may be actuated (e.g., rotated) to the first position to connect the first support member contact **174** with the first conductor **170**, which may provide an electrical connection from the first conductor **170** to the first load contact **162**. Additionally or alternatively, actuating the switch assembly **180** to the first position while the support member **124** is in the first configuration may connect the second support member contact **176** with the second conductor **172**, which may provide an electrical connection between the second conductor **172** and the second load contact **164**. For example and without limitation, actuating the switch assembly **180** to the first position when the support member **124** is in the first configuration may provide power from the pair of tracks **122** and/or a power source **190** to the electrical load **160**.

(83) With embodiments, such as generally illustrated in FIG. **12**, if a support member **124'** is connected with the tracks **130, 132** in a second configuration, the first side **124A** of the support member **124'** may be disposed proximate the second track **132**, and the second side **124B** of the support member **124'** may be disposed proximate the first track **130**. In the second configuration, the switch assembly **180** may be actuated (e.g., rotated by a user) to the second position to connect the first support member contact **174** with the first conductor **170**, which may provide an electrical connection from the first conductor **170** to the first load contact **162**. Additionally or alternatively, actuating the switch assembly **180** to the second position when the support member **124'** is in the second configuration may connect the second support member contact **176** with the second conductor **172**, which may provide an electrical connection between the second conductor **172** and the second load contact **164**. For example and without limitation, actuating the switch assembly **180** to the second position when the support member **124'** is in the second configuration may provide power from the pair of tracks **122** and/or a power source **190** to an electrical load **160**.

(84) In embodiments, such as generally illustrated in FIG. **12**, if a support member **124, 124'** is connected with the tracks **130, 132** in the first configuration and the switch assembly **180** is actuated to the second position, or if the support member **124, 124'** is connected with the tracks **130, 132** in the second configuration and the switch assembly **180** is actuated to the first position, an improper current may flow (e.g., a reverse polarity current) between the first conductor **170** and the second load contact **164** and/or between the second conductor **172** and the first load contact **162** may be provided. The electrical track assembly **120** and/or the switch assembly **180** may be configured to prevent such a current. For example and without limitation, the switch assembly **180** may include a blocking element **222**, such as a diode **202**, that may allow current to flow from the first support member contact **174** to the load **160** and to the second support member contact **176**,

but may prevent current flow from the second support member contact **176** to the load **160** and to the first support member contact **174**. The diode **202** may, for example, interrupt a current path that would otherwise flow from the first conductor **170** to the second support member contact **176**, to the second load contact **164**, to the first load contact **162**, to the first support member contact **174**, to the second conductor **172**.

(85) In embodiments, a switch assembly **180** may be configured to be switched toward the first side **142** of the mounting surface **140** into the first switch position when the support member **124**, **124'** is in the first configuration to connect the first load contact **162** with the first conductor **170**.

Additionally or alternatively, the switch assembly **180** may be configured to be switched toward the first side **142** of the mounting surface **140** into the second switch position when the support member **124**, **124'** is in the second configuration to electrically connect the first load contact **162** with the first conductor **170**. For example and without limitation, the switch assembly **180** may be actuated toward the first side **142** regardless of the configuration of the support member **124**, **124'** to provide a proper electrical connection. A label **200** (or other marking) may be disposed proximate the switch assembly **180** to indicate that the switch assembly **180** (and/or the first support member contact **174**) should be actuated toward the first side **142**. If the switch assembly **180** is actuated toward the second side **144**, power may not be provided to the electrical load **160**.

(86) With embodiments, such as generally illustrated in FIGS. **12-14**, a switch assembly **180** may include an indicator circuit **210**. The indicator circuit **210** may be configured to provide an indication to a user about whether the switch assembly **180** has been actuated correctly. The indication may include a visual indication and/or an audible indication, among others. An indicator circuit **210** may, for example, include a first light emitting diode (LED) **212** and/or a second LED **214**. The first LED **212** and the second LED **214** may be disposed in parallel with each other and/or with the diode **202** and the load **160** such that the diode **202** does not prevent current flow to the LEDs **212**, **214** and the indicator circuit **210** does not provide power to the load **160**. The first LED **212** may be disposed to allow current to flow from the first support member contact **174** to the second support member contact **176**, and in the presence of such current, may provide a first light (e.g., a green light), which may indicate correct actuation of the switch assembly **180** (see, e.g., FIG. **12**). The second LED **214** may be disposed to allow current to flow from the second support member contact **176** to the first support member contact **174**, and in the presence of such current, may provide a second light (e.g., a red light), which may indicate incorrect actuation of the switch assembly **180** (see, e.g., FIG. **13**). The indicator circuit **210** may include a resistor **216** upstream of the first LED **212** and downstream of the second LED **214**. The indicator circuit **210** may move with the switch assembly **180**, such as to the first position, the second position, and/or the third position.

(87) In embodiments, such as generally illustrated in FIGS. **15** and **16**, an electrical track assembly **120** may include a switch assembly **280** that may include a switch **282** (e.g., a manual electrical switch) that may be connected (e.g., electrically) to the first support member contact **174** and the second support member contact **176**. The manual electrical switch **282** may selectively connect the first support member contact **174** with the first load contact **162** and connect the second support member contact **176** with the second load contact **164**. A manual electrical switch **282** may include a first switch contact **284.sub.1**, a second switch contact **284.sub.2**, a third switch contact **284.sub.3**, a fourth switch contact **284.sub.4**, a fifth switch contact **284.sub.5**, and/or a sixth switch contact **284.sub.6**. The first switch contact **284.sub.1** may be connected (e.g., permanently, electrically) to the fourth switch contact **284.sub.4**. The second switch contact **284.sub.2** may be connected (e.g., permanently, electrically) to the fifth switch contact **284.sub.5**. The third switch contact **284.sub.3** may be connected (e.g., permanently, electrically) to the sixth switch contact **284.sub.6**. The fourth switch contact **284.sub.4** and the sixth switch contact **284.sub.6** may be connected (e.g., permanently, electrically) together and/or to first load contact **162** of the electrical load **160**. The fifth switch contact **284.sub.5** may be connected (e.g., permanently, electrically) to

the second load contact **164** of the electrical load **160**.

(88) With embodiments, a manual electrical switch **282** may move between a first position (see, e.g., FIG. **15**) and a second position (see, e.g., FIG. **16**). In the first position of the switch **282**, the first switch contact **284.sub.1** may be connected with the first support member contact **174**, the second switch contact **284.sub.2** may be connected to the second support member contact **176**, and/or the third switch contact **284.sub.3** may be open. If the support member **124, 124'** is in the first configuration and the switch **282** is in the first position (see, e.g., switch **282** of support member **124** of FIG. **15**), the switch **282** may provide an electrical connection from the first conductor **170** to the first load contact **162** and from the second load contact **164** to the second conductor **172**, which may provide power to the electrical load **160**. In the second position of the switch **282**, the second switch contact **284.sub.2** may be electrically connected to the first support member contact **174**, the third switch contact **284.sub.3** may be electrically connected to the second support member contact **176**, and the first switch contact **284.sub.1** may be open. If the support member **124, 124'** is in the second configuration and the switch **282** is in the second position (see, e.g., switch **282** of support member **124'** of FIG. **16**), the switch **282** may provide an electrical connection from the first conductor **170** to the first load contact **162** (e.g., via the second support member contact **176**, the third switch contact **284.sub.3**, and the sixth switch contact **284.sub.6**) and from the second load contact **164** to the second conductor **172** (e.g., via the fifth switch contact **284.sub.5**, the second switch contact **284.sub.2**, and the first support member contact **174**), which may provide power to the electrical load **160**. For example and without limitation, the switch **282** may be actuated to provide power to the electrical load **160** in either the first configuration or the second configuration of the support member **124, 124'**.

(89) In embodiments, if the switch **282** is in the second position when the support member **124, 124'** is in the first configuration (see, e.g., switch **282** of support member **124** of FIG. **16**) or if the switch **282** is in the first position when the support member **124, 124'** is in the second configuration (see, e.g., switch **282** of support member **124'** of FIG. **15**), flow of an improper current between the conductors **170, 172** and the electrical load **160** could result (e.g., a reversed polarity connection/flow). The switch assembly **180** may be configured to prevent such an improper flow of current, such as via a blocking element **222**. A blocking element **222** may, for example and without limitation, include electrical blocking elements, such as a first switch diode **286** and/or a second switch diode **288**, and/or include mechanical blocking elements, such as a block and/or lever. The first switch diode **286** may be connected to the fourth switch contact **284.sub.4** and may be configured to allow current flow from the fourth switch contact **284.sub.4** to the first load contact **162**, but may prevent current flow from the first load contact **162** to the fourth switch contact **284.sub.4**. The second switch diode **288** may be connected to the sixth switch contact **284.sub.6** and may be configured to allow current flow from the sixth switch contact **284.sub.6** to the first load contact **162**, but may prevent current flow from the first load contact **162** to the sixth switch contact **284.sub.6**. For example and without limitation, the first switch diode **286** and the second switch diode **288** may interrupt a current path from the first conductor **170** to the second conductor **172** when the switch **282** is in first position and the support member **124, 124'** is in the second configuration and when the switch **282** is in the second position and the support member **124, 124'** is in the first configuration, respectively.

(90) With embodiments, a switch **282** of a switch assembly **280** may be configured such that the correct position of the switch **282** is proximate the same side of a mounting surface **140** when the support member **124, 124'** is in the first configuration and when the support member **124, 124'** is in the second configuration. For example and without limitation, a switch **282** may be configured such that actuating the switch **282** toward the second side **144** (e.g., the left side of a vehicle) may provide power to the electrical load **160**. A label **300** (or other marking) may be disposed proximate the switch assembly **280** to indicate that the switch **282** should be actuated toward the second side **144**. If the switch **282** is actuated toward the first side **142** (e.g., a right side), power may not be

provided to the electrical load **160** (e.g., may be prevented by a switch diode **286, 288**).

(91) In embodiments, a first track **130** may be disposed proximate a first side **142** of a mounting surface and a second track **132** may be disposed proximate a second side **144** of the mounting surface **140**. A manual electrical switch **282** may be configured to be switched toward the first side **142** into a first switch position when the support member **124, 124'** is in the first configuration to connect (e.g., electrically) the first load contact **162** with the first conductor **170**. Additionally or alternatively, the manual electrical switch **282** may be configured to be switched toward the first side **142** into a second switch position when the support member **124, 124'** is in the second configuration to connect (e.g., electrically) the first load contact **162** with the first conductor **170**. The first switch position and the second switch position may be different and/or opposite. With embodiments, a switch assembly **280** may include an indicator circuit **210**, such as generally described above in connection with switch assembly **180**.

(92) With embodiments, such as generally illustrated in FIGS. **17** and **18**, an electrical track assembly **120** may include a switch assembly **380** that may include a manual electrical switch **382**. The manual electrical switch **382** may include a first switch contact **384.sub.1**, a second switch contact **384.sub.2**, a third switch contact **384.sub.3**, a fourth switch contact **384.sub.4**, a fifth switch contact **384.sub.5**, and/or a sixth switch contact **384.sub.6**. The first switch contact **384.sub.1** and/or the sixth switch contact **384.sub.6** may be connected (e.g., permanently, electrically) to the first support member contact **174**. The second switch contact **384.sub.2** may be connected (e.g., permanently, electrically) to the first load contact **162** of the electrical load **160**. The third switch contact **384.sub.3** and/or the fourth switch contact **384.sub.4** may be connected (e.g., permanently, electrically) to the second support member contact **176**. The fifth switch contact **384.sub.5** may be connected (e.g., permanently, electrically) to the second load contact **164**.

(93) In embodiments, a manual electrical switch **382** may include a first position (see, e.g., switch **382** of support member **124** in FIGS. **17** and **18**) in which the manual electrical switch **382** may electrically connect the first switch contact **384.sub.1** with the second switch contact **384.sub.2**, which may electrically connect the first support member contact **174** with the first load contact **162**, and/or the manual electrical switch **382** may connect the fourth switch contact **384.sub.4** with the fifth switch contact **384.sub.5**, which may connect the second support member contact **176** with the second load contact **164**. If the support member **124, 124'** is in the first configuration and the manual electrical switch **382** is in the first position, the first support member contact **174** may be in contact with the first conductor **170**, and the second support member contact **176** may be in contact with the second conductor **172**, which may provide power to the electrical load **160**.

(94) In embodiments, a manual electrical switch **382** may include a second position (see, e.g., switches **382** of support members **124'** of FIGS. **17** and **18**) that may connect the second switch contact **384.sub.2** with the third switch contact **384.sub.3**, which may electrically connect the second support member contact **176** with the first load contact **162**, and/or the manual electrical switch **382** may connect the fifth switch contact **384.sub.5** with the sixth switch contact **384.sub.6**, which may electrically connect the first support member contact **174** with the second load contact **164**. If the support member **124, 124'** is in the second configuration and the manual electrical switch **382** is in the second position, the second support member contact **176** may be in contact with the first conductor **170**, and the first support member contact **174** may be in contact with the second conductor **172**, which may provide power to the electrical load **160**.

(95) With embodiments, if a manual electrical switch **382** is in the first position when the support member **124, 124'** is in the second configuration and/or is in the second position when the support member **124, 124'** is in the first configuration (e.g., the opposite arrangements of those in FIGS. **17** and **18**), an improper flow of current (e.g., reversed polarity) could result. An electrical track assembly **120** and/or a switch assembly **380** may be configured to prevent such an improper flow of current. For example and without limitation, a diode **386** may be connected to the second switch contact **384.sub.2** (e.g., between the second switch contact **284.sub.2** and the first load contact

162). The diode **386** may be configured to allow current flow from the second switch contact **384.sub.2** to the first load contact **162**, but may prevent current flow from the first load contact **162** to the second switch contact **384.sub.2** (e.g., prevent a reversed polarity connection).

(96) In embodiments, a switch **382** may include a third position (e.g., a neutral position) in which none of the contacts **384.sub.1-6** are connected together. A switch **382** may, for example and without limitation, include a double pole double throw switch.

(97) With embodiments, a switch **382** of a switch assembly **380** may be configured such that the correct position of the switch **382** is proximate the same end of a mounting surface **140** when the support member **124, 124'** is in the first configuration and when the support member **124, 124'** is in the second configuration. For example and without limitation, a switch **382** may be configured such that actuating the switch **382** toward the first end **146** (e.g., a vehicle front) may provide power to the electrical load **160** regardless of the configuration of the support member **124, 124'**. A label **400** (or other marking) may be disposed proximate the switch assembly **380** to indicate that the switch **382** should be actuated toward the first end **146**. If the switch **382** is actuated toward the second end **148**, power may not be provided to the electrical load **160** (e.g., from the power source **190**).

(98) With embodiments, such as generally illustrated in FIG. **18**, a switch assembly **380** may include an indicator circuit **390** that may be configured in a similar manner as the indicator circuit **210** of switch assembly **180**. The indicator circuit **390** may be connected in parallel with the diode **386** and may include two LEDs **392, 394** disposed in opposite directions to provide an indication of whether the switch **382** has been actuated to the correct position. A first LED **392** may, for example, provide green light when the switch **382** is actuated to the correct position (e.g., to electrically connect the first load contact **162** with the first conductor **170** and to electrically connect the second load contact **164** with the second support member conductor **172**). A second LED **394** may, for example, provide red light when the switch **382** has been actuated to an incorrect position (e.g., that would otherwise connect the first load contact **162** with the second conductor **172** or connect the second load contact **164** with the first conductor **170**). The indicator circuit **390** may be connected to the second switch contact **384.sub.2**, the fifth switch contact **384.sub.5**, and/or the second load contact **164**. A resistor **396** may be disposed upstream of the first LED **392** and/or downstream of the second LED **394**.

(99) In embodiments, such as generally illustrated in FIGS. **15-18**, a first support member contact **174** and/or a second support member contact **176** of a support member **124, 124'** may move independently of the switch assembly **380** (or not move if the switch assembly **380** moves). For example and without limitation, the first support member contact **174** and the second support member contact **176** may move from a disengaged position in which the support member contacts **174, 176** do not contact the conductors **170, 172** to an engaged position in which the support member contacts **174, 176** do contact the conductors **170, 172**, regardless of the position of the switch assembly **380**. In the disengaged position, the support member contacts **174, 176** may, for example, be generally aligned with a longitudinal direction of the tracks **130, 132** (e.g., an X-direction) and may not restrict removal of the support member **124, 124'** from the tracks **130, 132** (see, e.g., hidden lines in FIGS. **9** and **10**). In the engaged position, the support member contacts **174, 176** may, for example, be generally aligned with a transverse direction (e.g., a Y-direction) and may restrict removal of the support member **124, 124'** from the tracks **130, 132** (e.g., may overlap with portions of the tracks **130, 132** in a vertical/Z-direction).

(100) With embodiments, a first support member contact **174** may be configured for selective electrical connection with the first conductor **170** when the support member **124, 124'** is connected to the pair of tracks **122** in the first configuration and/or for selective electrical connection with the second conductor **172** when the support member **124, 124'** is connected with the pair of tracks **122** in the second configuration. Additionally or alternatively, a second support member contact **176** may be configured for selective electrical connection with the second conductor **172** when the

support member **124**, **124'** is connected to the pair of tracks **122** in the first configuration and/or selective electrical connection with the first conductor **170** when the support member **124**, **124'** is connected with the pair of tracks **122** in the second configuration.

(101) In embodiments, such as generally illustrated in FIGS. **12-18**, an electrical track assembly **120** may include a support member **124** and a second support member **124'** (and/or additional support members). The second support member **124'** may include the same or a similar configuration as the support member **124**, and may be configured for connection with, removal from, and movement along the first track **130** and the second track **132**. The second support member **124'** may include a first support member contact **174** (e.g., a third support member contact of the electrical track assembly **120**) that may be configured for selective electrical connection with the first conductor **170** when the second support member **124'** is connected to the first track **130** and/or the second track **132** in the first configuration and with the second conductor **172** when the second support member **124'** is connected with the first track **130** and/or the second track **132** in the second configuration. The second support member **124'** may include a second support member contact **176** (e.g., a fourth support member contact of the electrical track assembly **120**) that may be configured for selective electrical connection with the second conductor **172** when the second support member **124'** is connected to the first track **130** and/or the second track **132** in the first configuration and with the first conductor **170** when the second support member **124'** is connected with the first track **130** and/or the second track **132** in the second configuration. The second support member **124'** may include an electrical load **160** (e.g., a second electrical load) including a first load contact **162** and a second load contact **164** (e.g., third and fourth load contacts of the electrical assembly **20**), and/or may include a switch assembly **180**, **280**, **380** (e.g., a second switch assembly of the electrical assembly **20**) that may be configured to selectively electrically connect the first conductor **170** with the first load contact **162**, selectively electrically connect the second conductor **172** with the second load contact **164**, prevent current flow between the first conductor **170** and the second load contact **164**, and/or prevent current flow between the second conductor **172** and the first load contact **162**. The first switch assembly **180**, **280**, **380** of the support member **124** may include a first manual electrical switch **282**, **382**, and the (second) switch assembly **180**, **280**, **380** of the second support member **124'** may include a (second) manual electrical switch **282**, **382** connected to the first support member contact **174**, the second support member contact **176**, the first load contact **162**, and/or the second load contact **164** of the support member **124'**. The second switch assembly **180**, **280**, **380** may include a second diode **202**, **286**, **288**, **386** that may be connected to the second manual electrical switch **282**, **382**.

(102) With embodiments, a support member **124** and a second support member **124'** may be configured for connection with the pair of tracks **122** at the same time with some or all of the following arrangements: (i) both the first support member **124** and the second support member **124'** in the first configuration, (ii) the first support member **124** in the first configuration and the second support member **124'** in the second configuration, (iii) the first support member **124** in the second configuration and the second support member **124'** in the first configuration, and/or (iv) the first support member **124** and the second support member **124'** in the second configuration. In some or each of such arrangements, a switch assembly **180**, **280**, **380** may be configured to facilitate current flow between (i) the first load contact **162** of the first support member **124** and the first conductor **170**, (ii) the second load contact **164** of the first support member **124** and the second conductor **172**, (iii) the first load contact **162** of the second support member **124'** (e.g., a third load contact) and the first conductor **170**, and/or (iv) the second load contact **164** of the second support member **124'** (e.g., a fourth load contact) and the second conductor **172**.

(103) With embodiments, such as generally illustrated in FIG. **19**, an electrical track assembly **120** may include a first pair of tracks **122** and a second pair of tracks **122'**. One or more support members (e.g., support members **124**, **124'**) may be connected with the first pair of tracks **122** and one or more support members (e.g., support members **424**, **524**) may be connected to the second

pair of tracks **122'**. One or more support members (e.g., support member **624**) may be connected to one or more tracks of both the first pair of tracks **122** and the second pair of tracks **122'**. Some or all of the support members **124**, **124'**, **424**, **524**, **624** may include a respective switch assembly **180**, **280**, **380**, which may allow for the support members to be connected with the tracks **122**, **122'** in multiple configurations and provide power to a respective electrical load **160** (e.g., with the proper polarity).

(104) With embodiments, a support member **124**, **124'**, **424**, **524**, **624** may include, be connected to, and/or support one or more of a variety of components. For example and without limitation, as generally illustrated in FIGS. **8** and **19**, a support member **124**, **124'**, **424**, **524** may include, be connected to, and/or support a seat **126**, such as a vehicle seat. Additionally or alternatively, a support member (e.g., support member **624**) may include, be connected to, and/or support a table/console **128**.

(105) In embodiments, such as generally illustrated in FIGS. **11**, **20**, **21**, and **22**, an electrical track assembly **120** may include one or more asymmetric features **220**, **320**. The one or more asymmetric features **220**, **320** may include one or more features (e.g., protrusions, tracks, bars, etc.) that may be configured to facilitate preventing an improper electrical connection. For example and without limitation, asymmetric feature(s) **220** may be configured to mechanically prevent connection of the first support member contact **174** with the second conductor **172** and/or connection of the second support member contact **176** with the first conductor **170** (see, e.g., FIG. **11**). The feature(s) **220** may be included with and/or connected to the first track **130** and may restrict movement (e.g., rotation) of the second support member contact **176** into engagement with the first conductor **170**.

(106) With embodiments, such as generally illustrated in FIGS. **11**, **20**, **21**, and **22**, a support member **124**, **124'** and/or a switch assembly **180**, **280**, **380** may include a blocking element **222**, **322** that may be configured to engage and/or detect the asymmetric features **220**. A blocking element **222**, **322** may, for example, be configured to detect/engage an asymmetric feature **220**, **320** to prevent an improper electrical connection. For example and without limitation, a blocking element **322** may detect/engage an asymmetric feature **220** to prevent electrical connection of the second support member contact **176** with the first conductor **170** (see, e.g., FIG. **11**). The blocking element **222** may be connected to and/or incorporated with the second support member contact **176** such that the blocking element **222** moves with the second support member contact **176** and contacts the feature **220** to restrict movement of the second support member contact **176**.

(107) In embodiments, such as generally illustrated in FIGS. **20**, **21**, and **22**, a blocking element **322** may be configured to limit actuation of one or more switches of a switch assembly. For example and without limitation, a blocking element **322** of a switch assembly **380** may be configured to physically/mechanically restrict and/or prevent actuation of a switch **382** of the switch assembly **380** into a position that would result in an improper connection (e.g., reversed polarity). The blocking element **322** may include and/or be configured as a linkage that may include a first portion **324**, a second portion **326**, and/or a third portion **328**, and/or that may be configured to restrict actuation of a switch assembly **380**. The first portion **324** may, for example, include a lever (e.g., bar, rod, tube, elongated plate, etc.) that may extend substantially in a Z-direction (e.g., in a vertical direction). The second portion **326** may, for example, include a lever that may extend substantially in a Z-direction in parallel with and offset from the first portion **324**, such as in a Y-direction (e.g., if the switch **382** is substantially aligned with the Y-direction, such as generally shown in FIGS. **20-22**) and/or in an X-direction (e.g., if the switch **382** is substantially aligned with an X-direction).

(108) With embodiments, a third portion **328** may include a lever that may be rotatably connected with the support member **124**, **124'**, such as via a pivot **330**. The third portion **328** may rotate in a plane substantially parallel with the switch **382**. For example and without limitation, if the switch **382** is substantially aligned with the Y-direction, such as generally shown in FIGS. **20-22**, the third portion may, for example, be configured to rotate substantially in a Y-Z plane, and if the switch **382**

is substantially aligned with an X-direction, the third portion may, for example, be configured to rotate substantially in a X-Z plane.

(109) In embodiments, the first portion **324** may be connected (e.g., rotatably) at or about a first end of the third portion **328**, and/or the second portion **326** may be connected (e.g., rotatably) at or about a second end of the third portion **328**, which may be opposite the first end. The portions **324**, **326**, **328** may be configured such that movement (e.g., translation) of the first portion **324** in a vertical direction may cause rotation of the third portion **328**, which may cause movement (e.g., translation) of the second portion **326** in a substantially opposite vertical direction.

(110) In embodiments, the first portion **324** may include a first portion blocking position (see, e.g., FIG. **20**), and/or the second portion **326** may include a second portion blocking position (see, e.g., FIG. **21**). In the first portion blocking position, the first portion **324** may be disposed to restrict and/or prevent movement of a lever **382A** of the switch **382** to the second switch position. For example, in the first portion blocking position, the first portion **324** may be disposed adjacent to the lever **382A** (e.g., to the left of the lever **382A** in FIG. **20**), which may restrict and/or prevent the lever **382A** from moving to the second position (e.g., to the left in FIG. **20**), which may prevent an improper electrical connection between the load **160** and the power source **190**. In the first portion blocking position, a top of the first portion **324** may be disposed at a higher vertical level than a top of the second portion **326**, and the second portion **326** may be in a second portion inactive position.

(111) With embodiments, in the second portion blocking position, the second portion **326** may be disposed to restrict and/or prevent movement of the lever **382A** of the switch **382** to the first switch position. For example, in the second portion blocking position, the second portion **326** may be disposed adjacent to the lever **382A** (e.g., to the left of the lever **382A** in FIG. **21**), which may restrict and/or prevent the lever **382A** from moving to the first position (e.g., to the left in FIG. **21**), which may prevent an improper electrical connection between the load **160** and the power source **190**. In the second portion blocking position, the top of the second portion **326** may be disposed at a higher vertical level than the top of the first portion **324**, and the first portion **324** may be in a first portion inactive position.

(112) With embodiments, the first portion **324** may extend (e.g., downward) from the support member **124**, **124'** such that when the support member **124**, **124'** is disposed on the tracks **130**, **132**, the first portion **324** may extend into one of the tracks **130**, **132**, such as to detect/engage an asymmetric feature **320**, if present. For example, if the support member **124**, **124'** is disposed on the tracks **130**, **132** such that the first connection portion **150** is connected with and/or inserted into the first track **130**, the first portion **324** may extend down into the first track **130** and engage the asymmetric feature **320**. Engagement between the first portion **324** and the asymmetric feature **320** may cause the first portion **324** to translate upward from a first portion inactive position into the first portion blocking position (e.g., automatically), which may cause rotation of the third portion **328** (e.g., clockwise in FIG. **20**), which may cause the second portion **326** to translate downward out of a second portion blocking position and into a second portion inactive position.

(113) In embodiments, a blocking element **322** may be biased such that the second portion **326** is biased toward the second portion blocking position and/or such that the first portion **324** is biased toward the first portion inactive position. The blocking element **322** may, for example, be biased by gravity acting on the first portion **324**, which may be heavier and/or connected at a greater distance from the pivot **330** than the second portion **326**, and/or may be biased by a biasing member that may be included with and/or connected to the pivot **330** (e.g., the pivot **330** may bias the third portion counterclockwise in FIG. **20** and clockwise in FIG. **21**). For example and without limitation, if the support member **124**, **124'** is disposed on the tracks **130**, **132** such that the first connection portion **150** is connected with and/or inserted into the second track **132** (see, e.g., FIG. **21**), the first portion **324** may extend down into the first track **132**, but there may not be an asymmetric feature to engage, so the blocking element **322** may remain in the same state, which may include the first portion **324** in the first portion inactive position and the second portion **326** in

the second blocking position.

(114) If the first track **130** includes an asymmetric feature **220, 320**, the second track **132** may not include an asymmetric feature **220, 320** or may include a different type/configuration of an asymmetric feature. An electrical track assembly **120** may include an opposite configuration in which the second track **132** may include the asymmetric feature **220, 320**, and a blocking element **222, 322** may detect/engage an asymmetric feature **220, 320** of the second track **132** to prevent an improper electrical connection.

(115) In some circumstances, it may be desirable for an electrical track assembly to be configured for connections between support members and tracks without electrical switching, such as automatic switching via a control circuit **52** or via a switch assembly **180, 280, 380**.

(116) With embodiments, such as generally illustrated in FIGS. **23A-23C**, an electrical track assembly **1120** may include a first track/rail **1130**, a second track/rail **1132**, and/or an electrical load **1160** that may be connected to a support member **1124**. An electrical track assembly **1120** may include one or more conductors **1170, 1170.sub.2, 1172, 1172.sub.2** (e.g., electrical bus bars) that may be connected to and/or included with the first track/rail **1130** and/or the second track/rail **1132**. For example, in some configurations, a first track **1130** may include a first conductor **1170** and/or a second conductor **1172**, and the second track **1132** may not include a conductor (see, e.g., FIGS. **23A** and **24**). As generally illustrated in FIG. **24**, for example, a support member **1124** may include a first sets of contacts **1174, 1176** for electrically contacting the conductors **1170, 1172** when a support member **1124** is in a first orientation while the second set of contacts **1174.sub.2, 1176.sub.2** may not provide an electrical connection. In the second orientation of the support member **1124**, the second set of contacts **1174.sub.2, 1176.sub.2** may be configured to electrically contact conductors **1170, 1172** while the first set of contacts **1174, 1176** may not provide an electrical connection. For example, in a first orientation, the first set of contacts **1174, 1176** may move into electrical contact with conductors **1170, 1172**, and, in the second orientation, the second set of contacts **1174.sub.2, 1176.sub.2** may move into electrical contact with conductors **1170, 1172**. With such a configuration, the support member **1124** may automatically electrically connect with the conductors **1170, 1172** of the track **1130** regardless of the orientation of the support member **1124**.

(117) In other configurations, for example, a first track **1130** may include a first conductor **1170** and/or a second conductor **1172**, and the second track **1132** may include an additional conductor **1170.sub.2** or **1172.sub.2** (see, e.g., FIGS. **23B, 25, and 26**). Similar to the configuration shown in FIG. **24**, a support member **1124** may include a first set of contacts **1174, 1176** and a second set of contacts **1174.sub.2, 1176.sub.2** that may be configured to electrically contact conductors **1170, 1172, 1170.sub.2** of the tracks **1130, 1132** (see, e.g., FIG. **25**). In the first orientation, the first set of contacts **1174, 1176** may electrically contact the conductors **1170, 1172**, respectively of the first track **1130**, and the third contact **1174.sub.2** may contact the conductor **1170.sub.2** of the second track **1132**. The first conductor **1170** and the third conductor **1170.sub.2** may be electrically connected to positive contacts **192, 196** of a power source **190**, and the second conductor **1172** may be electrically connected with a negative contact **194** of the power source **190**.

(118) In another configuration, the second track **1132** may include a fourth conductor **1172.sub.2** (e.g., instead of a third conductor **1170.sub.2**), and the second contact **1176** and the fourth contact **1176.sub.2** may be configured to electrically contact the fourth conductor **1172.sub.2** in the second and first orientations, respectively (see, e.g., FIG. **26**). In the first orientation, as shown in FIG. **26**, the third contact **1174.sub.2** may not be electrically connected to a conductor, and, in the second orientation, the first contact **1174** may not be electrically connected to a conductor.

(119) With other configurations, for example, a first track **1130** may include a first conductor **1170** and a second conductor **1172**, and the second track **1132** may include a third conductor **1170.sub.2** and a fourth conductor **1172.sub.2** (see, e.g., FIGS. **23C, 27, 28A, and 28B**). A power source **190** may be connected to the four conductors **1170, 1172, 1170.sub.2, 1172.sub.2**. For example, the

second conductor **1172** and the fourth conductor **1172.sub.2** may be connected to a negative or ground contact **194** of the power source, the first conductor **1170** may be connected to a first positive contact of the power source **190**, and/or the third conductor **1170.sub.2** may be connected to a second positive contact of the power source **190**. The power source **190** may, in some examples, provide power with different characteristics (e.g., current, voltage, etc.) to the first and second positive contacts **192**, **196**, which may be utilized by the support member **1124**, the electrical track assembly **1120**, and/or a vehicle **100** to determine the orientation in which the support member **1124** is disposed.

(120) As generally illustrated in FIG. **27**, in some configurations, a support member **1124** may include two sets of electrical contacts **1174**, **1176** and **1174.sub.2**, **1176.sub.2** such that a contact electrically contacts/connects with a respective one of the conductors **1170**, **1172**, **1170.sub.2**, **1172.sub.2** in both orientations of the support member **1124**.

(121) In another configuration, such as generally shown in FIGS. **28A** and **28B**, a support member **1124** may include a single set of contacts **1174**, **1176** that electrically contact a first of conductors **1170**, **1172** in the first orientation (see, e.g., FIG. **28A**), and electrically contact a second set of conductors **1170.sub.2**, **1172.sub.2** of the second track **1132** in the second orientation (see, e.g., FIG. **28B**). In the first orientation, the third and fourth conductors **1170.sub.2**, **1172.sub.2** may not be electrically connected (e.g., directly) with electrical contacts of the support member **1124**. In the second orientation, the first and second conductors **1170**, **1172** may not be electrically connected (e.g., directly) with electrical contacts of the support member **1124**.

(122) In embodiments, an electrical track assembly **120** may be configured to provide power to an electrical load **160** of a support member **124**, **124'**, **424**, **524**, **624** via manual and/or mechanical operation of a switch assembly **180**, **280**, **380**, such as without utilizing an electronic controller (e.g., without an ECU **28**) and/or utilizing an automatic control circuit (e.g., without a control circuit **52**) to control or actuate any portion of the switch assembly **180**, **280**, **380**. Such a configuration may reduce complexity and/or power consumption by the electrical track assembly **120** compared to designs in which an electronic controller or automatic control circuit is utilized, such as in one or more embodiments of electrical assembly **20**.

(123) In embodiments, such as generally illustrated in FIG. **29**, an electrical track assembly **2120** that may include a first track/rail **2130** and a second track/rail **2132**. The first track **2130** may include a first conductor **2170** and a second conductor **2172** that may be disposed in a stacked configuration (e.g., with the first conductor **2170** above the second conductor **2172**) and/or that may extend in the X-direction, such as along some or most of the first track **2130**. The second track **2132** may include a third conductor **2170.sub.2** and a fourth conductor **2172.sub.2** that may be disposed in a stacked configuration (e.g., with the fourth conductor **2172.sub.2** above the third conductor **2170.sub.2**) and/or that may extend in the X-direction, such as along some or most of the second track **2132**. A support member **2124** may be configured for selective connection (e.g., mechanical and electrical) with, movement along, and/or removal from the tracks **2130**, **2132**. The support member **2124** may include a first electrical contact **2174**, a second electrical contact **2176**, a third electrical contact **2174.sub.2**, and/or a fourth electrical contact **2176.sub.2**. In a first connected configuration, the first and second electrical contact **2174**, **2176** may move (e.g., rotate) into electrical contact with first and second conductors **2170**, **2172**, respectively, and the third and fourth contacts **2174.sub.2**, **2176.sub.2** may move into contact with the fourth and third conductors **2172.sub.2**, **2170.sub.2**, respectively. A control circuit **2152** of the support member **2124** may be configured to ensure that the first and fourth conductors **2170**, **2172.sub.2** of the tracks **2130**, **2132** are electrically connected with positive contact(s) **2162** of the electrical load **2160** and that the second and third conductors **2172**, **2170.sub.2** are electrically connected with the negative contact(s) **2164** of the electrical load **2160**. The control circuit **2152** may be configured in the same or a similar manner as the control circuit **52** described above.

(124) Various examples/embodiments are described herein for various apparatuses, systems, and/or

methods. Numerous specific details are set forth to provide a thorough understanding of the overall structure, function, manufacture, and use of the examples/embodiments as described in the specification and illustrated in the accompanying drawings. It will be understood by those skilled in the art, however, that the examples/embodiments may be practiced without such specific details. In other instances, well-known operations, components, and elements have not been described in detail so as not to obscure the examples/embodiments described in the specification. Those of ordinary skill in the art will understand that the examples/embodiments described and illustrated herein are non-limiting examples, and thus it can be appreciated that the specific structural and functional details disclosed herein may be representative and do not necessarily limit the scope of the embodiments.

(125) Reference throughout the specification to “examples,” “in examples,” “with examples,” “various embodiments,” “with embodiments,” “in embodiments,” or “an embodiment,” or the like, means that a particular feature, structure, or characteristic described in connection with the example/embodiment is included in at least one embodiment. Thus, appearances of the phrases “examples,” “in examples,” “with examples,” “in various embodiments,” “with embodiments,” “in embodiments,” or “an embodiment,” or the like, in places throughout the specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more examples/embodiments. Thus, the particular features, structures, or characteristics illustrated or described in connection with one embodiment/example may be combined, in whole or in part, with the features, structures, functions, and/or characteristics of one or more other embodiments/examples without limitation given that such combination is not illogical or non-functional. Moreover, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the scope thereof.

(126) It should be understood that references to a single element are not necessarily so limited and may include one or more of such element. Any directional references (e.g., plus, minus, upper, lower, upward, downward, left, right, leftward, rightward, top, bottom, above, below, vertical, horizontal, clockwise, and counterclockwise) are only used for identification purposes to aid the reader's understanding of the present disclosure, and do not create limitations, particularly as to the position, orientation, or use of examples/embodiments.

(127) Joinder references (e.g., attached, coupled, connected, and the like) are to be construed broadly and may include intermediate members between a connection of elements and relative movement between elements. As such, joinder references do not necessarily imply that two elements are directly connected/coupled and in fixed relation to each other. The use of “e.g.” in the specification is to be construed broadly and is used to provide non-limiting examples of embodiments of the disclosure, and the disclosure is not limited to such examples. Uses of “and” and “or” are to be construed broadly (e.g., to be treated as “and/or”). For example and without limitation, uses of “and” do not necessarily require all elements or features listed, and uses of “or” are inclusive unless such a construction would be illogical.

(128) While processes, systems, and methods may be described herein in connection with one or more steps in a particular sequence, it should be understood that such methods may be practiced with the steps in a different order, with certain steps performed simultaneously, with additional steps, and/or with certain described steps omitted.

(129) All matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative only and not limiting. Changes in detail or structure may be made without departing from the present disclosure.

Claims

1. A support member for an electrical track assembly, the support member comprising: an electrical load including a first load contact and a second load contact; and a switch assembly configured to: selectively provide electrical connection between a first conductor associated with a first track of a pair of tracks of the electrical track assembly and the first load contact, selectively provide electrical connection between a second conductor associated with a second track of the pair of tracks and the second load contact, and control current flow (i) between the first conductor and the second load contact and (ii) between the second conductor and the first load contact.
2. The support member of claim 1, further comprising: a first support member contact; and a second support member contact; wherein the switch assembly comprises a manual switch electrically connected to the first support member contact, the second support member contact, the first load contact, and the second load contact, the manual switch is configured to switch into a first switch position when the support member is in a first configuration to connect the first load contact with the first conductor, and the manual switch is configured to switch into a second switch position when the support member is in a second configuration to connect the first load contact with the first conductor.
3. The support member of claim 2, wherein the manual switch includes a third switch position in which the manual switch does not connect the first load contact with either of the first conductor or the second conductor.
4. The support member of claim 2, wherein, when the support member is in a first configuration, the first support member contact is configured to move between one of (i) an engaged position in which the first support member contact is in contact with the first conductor and restricts removal of the support member from the first track, or (ii) a disengaged position in which the first support member contact is not in contact with the first conductor and does not restrict removal of the support member from the first track.
5. The support member of claim 2, wherein in the first switch position, the manual switch connects the first support member contact with the first load contact, and in the second switch position, the manual switch connects the first support member contact with the second load contact.
6. The support member of claim 5, wherein the switch assembly further comprises an indicator circuit configured to: provide at least one of a first visual indication or an audible indication if (i) the manual switch is switched to the second switch position when the support member is in the first configuration, or (ii) the manual switch is switched to the first switch position when the support member is in the second configuration, and provide a second visual indication if (a) the manual switch is switched to the first switch position when the support member is in the first configuration, or (b) the manual switch is switched to the second switch position when the support member is in the second configuration.
7. The support member of claim 6, wherein the switch assembly further comprises a blocking element including a blocking diode configured to restrict current flow between the first load contact and the second conductor, the indicator circuit includes a pair of light emitting diodes disposed in parallel with each other and in parallel with at least one of a switch and the blocking diode, and the pair of light emitting diodes are connected in opposite directions.
8. The support member of claim 1, wherein the switch assembly includes a double pole double throw switch.
9. The support member of claim 1, wherein the switch assembly comprises a blocking element including a diode configured to restrict current flow between the first load contact and the second conductor.
10. An electrical track assembly, comprising: the support member of claim 1, wherein the support member further comprises a first support member contact and a second support member contact; the pair of tracks including: the first track, and the second track; and a second support member configured for connection with, removal from, and movement along the first track and the second

track, the second support member including: a third support member contact configured for selective electrical connection with the first conductor when the second support member is connected to the pair of tracks in a first configuration, and with the second conductor when the second support member is connected with the pair of tracks in a second configuration; a fourth support member contact configured for selective electrical connection with the second conductor when the second support member is connected to the pair of tracks in the first configuration, and with the first conductor when the second support member is connected with the pair of tracks in the second configuration; a second electrical load including a third load contact and a fourth load contact; and a second switch assembly configured to selectively connect the first conductor with the third load contact, selectively connect the second conductor with the fourth load contact, control current flow between the first conductor and the fourth load contact, and control current flow between the second conductor and the third load contact.

11. The electrical track assembly of claim 10, wherein the support member and the second support member are configured for connection with the pair of tracks at the same time with each of the following arrangements: (i) the support member and the second support member are in the first configuration, (ii) the support member in the first configuration and the second support member in the second configuration, (iii) the support member in the second configuration and the second support member in the first configuration, and (iv) the support member and the second support member in the second configuration, and wherein with each of the arrangements, the switch assembly is further configured to facilitate current flow (a) between the first conductor and the first load contact, and (b) between the first conductor and the third load contact.

12. The support member of claim 11, wherein the switch assembly is further configured to be moved by a user from a disengaged position in which the support member is not electrically connected with the first conductor and the second conductor, to an engaged position in which the support member is electrically connected with the first conductor and the second conductor.

13. An electrical track assembly, comprising: the support member of claim 1, the support member including a first support member contact and a second support member contact; the first track including the first conductor; and the second track including the second conductor; wherein the switch assembly is connected to or includes the first support member contact and the second support member contact.

14. The electrical track assembly of claim 13, wherein the switch assembly is further configured to move between a first switch assembly position when the support member is in a first configuration and a second switch assembly position when the support member is in a second configuration to connect the first support member contact to the first conductor and connect the second support member contact to the second conductor.

15. An electrical track assembly, comprising: the support member of claim 1; the first track including the first conductor; and the second track including the second conductor; wherein the first track includes an asymmetric feature.

16. The electrical track assembly of claim 15, wherein the switch assembly includes a blocking element configured to engage the asymmetric feature and mechanically restrict actuation of the switch assembly.

17. The electrical track assembly of claim 16, wherein the blocking element comprises a first portion, a second portion, and a third portion, and wherein the first portion is rotatably connected about a first end of the third portion, and the second portion is rotatably connected about a second end of the third portion, which is opposite to the first end, such that movement of the first portion in a vertical direction causes rotation of the third portion, which further causes movement of the second portion in a substantially opposite vertical direction.

18. An electrical track assembly, comprising: the support member of claim 1, wherein the support member further comprises a first support member contact and a second support member contact; and the pair of tracks, including: the first track including the first conductor, and the second track

including the second conductor, wherein the first support member contact is configured for selective electrical connection with the first conductor when the support member is connected to the pair of tracks in a first configuration, and with the second conductor when the support member is connected with the pair of tracks in a second configuration, and the second support member contact is configured for selective electrical connection with the second conductor when the support member is connected to the pair of tracks in the first configuration, and with the first conductor when the support member is connected with the pair of tracks in the second configuration.

19. An electrical track assembly, comprising: the support member of claim 1; and a pair of tracks, including: the first track including the first conductor, and the second track including the second conductor, wherein the first track is disposed proximate a first side of a mounting surface; and the second track is disposed proximate a second side of the mounting surface, and the switch assembly comprises a manual switch configured to: switch into a first switch position toward the first side when the support member is in a first configuration to connect the first load contact with the first conductor, and switch into a second switch position toward the first side when the support member is in a second configuration to connect the first load contact with the first conductor.

20. The electrical track assembly of claim 19, wherein a label is positioned proximate the first side of the manual switch to indicate that the manual switch is actuated toward the first side.

21. A support member for a track assembly, the support member comprising: a first support member contact configured for selective electrical connection with a first conductor of the track assembly in a first orientation of the support member and a second conductor of the track assembly in a second orientation of the support member; a second support member contact configured for selective electrical connection with said second conductor in the first orientation and the first conductor in the second orientation; an electrical load including a first load contact and a second load contact; and a switch assembly configured to: selectively provide current to the first support member contact and the second support member contact, and control current flow (i) between the first conductor and the second load contact and (ii) between the second conductor and the first load contact.

22. An electrical switch assembly, comprising: a manual switch configured to: selectively provide electrical connection between a first conductor associated with a track of an electrical track assembly, and a first load contact of an electrical load, selectively provide electrical connection between a second conductor associated with another track of the electrical track assembly, and a second load contact of the electrical load, and control current flow (i) between the first conductor and the second load contact and (ii) between the second conductor and the first load contact.
